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Semiconductor Manufacturing Apparatus, Pushup Unit, and Method for Manufacturing Semiconductor Device

Abstract

To provide a new mechanism for driving a plurality of blocks of a pushup unit. A pushup unit includes a mechanism section for independently moving each of a plurality of blocks up and down. The mechanism section includes a first mechanism section having: (a) a first member having an upper surface, a lower surface opposite the upper surface, and a through-hole penetrating between the upper surface and the lower surface; (b) a first rod connected to the upper surface of the first member to convey the vertical motion of the first member to the blocks; and (c) a second rod extended from below through the through-hole to convey the vertical motion to the blocks.

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Background/Summary

CLAIM OF PRIORITY

[0001] The present application claims priority from Japanese patent application JP 2024-032076 filed on Mar. 4, 2024, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

[0002] The present disclosure relates to a semiconductor manufacturing apparatus, and is applicable, for example, to a die bonder having a pushup unit.

[0003] One step in a semiconductor device manufacturing process is a peeling step in which a die separated from a wafer is peeled off from a dicing tape. In the peeling step, for example, the die is pushed up from the back surface of the dicing tape by a pushup unit, then peeled off one by one from the dicing tape held by a wafer supply unit, and picked up by using a suction nozzle such as a collet disposed on a pickup head or a bond head.

[0004] For example, the pushup unit removes the dicing tape from the periphery of the die by moving a plurality of blocks up and down. A drive section including a motor and a plunger mechanism may be provided for each block (refer, for example, to Japanese Unexamined Patent Application Publication No. 2017-224640).

SUMMARY OF THE INVENTION

[0005] An object of the present disclosure is to provide a new mechanism for driving a plurality of blocks of a pushup unit. Other objects and novel features will be apparent from the description of this document and the accompanying drawings.

[0006] A brief summary of representative aspects of the present disclosure is given below.

[0007] In short, the pushup unit includes a mechanism section for independently moving each of a plurality of blocks up and down. The mechanism section includes a first mechanism section having: (a) a first member having an upper surface, a lower surface opposite the upper surface, and a through-hole penetrating between the upper surface and the lower surface; (b) a first rod connected to the upper surface of the first member to convey the vertical motion of the first member to the blocks; and (c) a second rod extended from below through the through-hole to convey the vertical motion to the blocks.

[0008] The present disclosure makes it possible, for example, to increase the number of pushup blocks.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a schematic top view illustrating an example of the configuration of a die bonder according to an embodiment.

[0010] FIG. 2 is a diagram illustrating a schematic configuration as viewed in the direction of arrow A in FIG. 1.

[0011] FIG. 3 is a schematic cross-sectional view illustrating essential parts of a wafer supply unit depicted in FIG. 1.

[0012] FIG. 4 is a block diagram illustrating a schematic configuration of a control system of the die bonder depicted in FIG. 1.

[0013] FIG. 5 is a flowchart illustrating a method for manufacturing semiconductor device by using the die bonder depicted in FIG. 1.

[0014] FIG. 6 is a front view of a pushup unit in the embodiment.

[0015] FIG. 7 is a schematic diagram illustrating the pushup unit depicted in FIG. 6.

[0016] FIG. **8** is a top view of a head section depicted in FIG. **6**.

[0017] FIG. **9** is a longitudinal cross-sectional view of an upper plate section depicted in FIG. **6**.

[0018] FIG. **10** is a transverse cross-sectional view of the upper plate section taken along line C-C in FIG. **9**.

[0019] FIG. **11** is a transverse cross-sectional view of the upper plate section taken along line E-E in FIG. **9**.

[0020] FIG. **12** is a longitudinal cross-sectional view of a lower block section depicted in FIG. **6**.

[0021] FIG. **13** is a transverse cross-sectional view of the lower block section taken along line F-F in FIG. **12**.

[0022] FIG. **14** is a top view illustrating an example of layout of a drive section depicted in FIG. **6**.

[0023] FIG. **15** is a side view illustrating an example layout of the drive section depicted in FIG. **6**.

[0024] FIG. **16** is a diagram illustrating an example of the configuration of the drive section depicted in FIG. **15**.

[0025] FIG. **17** is a diagram illustrating another example of the configuration of the drive section depicted in FIG. **15**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0026] An embodiment will be described below with reference to the drawings. It should be noted that in the following description, the same components are indicated by the same reference numerals, and the repeated description thereof may be omitted. It should be noted that to make the description more clearly, the drawings may be schematically represented for the widths, thicknesses, shapes, and the like of the respective portions as compared with the actual form. In addition, also between a plurality of mutual drawings, the dimension relationships between the respective elements, the ratios between the respective elements, and the like do not always coincide with each other.

[0027] A configuration of a die bonder, which is an embodiment of a semiconductor manufacturing apparatus, will now be described with reference to FIGS. **1** to **3**. FIG. **1** is a schematic top view illustrating an example of the configuration of a die bonder according to an embodiment. FIG. **2** is a diagram illustrating a schematic configuration as viewed in the direction of arrow A in FIG. **1**. FIG. **3** is a schematic cross-sectional view illustrating essential parts of a wafer supply unit depicted in FIG. **1**.

[0028] The die bonder **1** broadly includes a wafer supply unit **10**, a pickup unit **20**, an intermediate stage unit **30**, a bonding unit **40**, a carrier unit **50**, a substrate supply unit **60**, a substrate unloading unit **70**, and a control unit (controller) **80**. The Y2-Y1 direction is the front-back direction of the die bonder **1**, the X2-X1 direction is the left-right direction, and the Z1-Z2 direction is the vertical direction. The wafer supply unit **10** is disposed on the front side of the die bonder **1**, and the bonding unit **40** is disposed on the back side.

[0029] The wafer supply unit **10** includes a wafer cassette lifter **11**, a wafer holding table **12**, a pushup unit **13**, and a wafer recognition camera **14**.

[0030] The wafer cassette lifter **11** vertically moves a wafer cassette (not depicted) to a wafer transfer height. A plurality of wafer rings WR are stored in the wafer cassette. A wafer correction chute (not depicted) aligns the wafer rings WR supplied from the wafer cassette lifter **11**. A wafer extractor (not depicted) removes the wafer rings WR from the wafer cassette, and supplies the wafer rings WR to the wafer holding table **12** or removes the wafer rings WR from the wafer holding table **12** and stores the wafer rings WR in the wafer cassette.

[0031] The wafer holding table **12** has an expand ring **121** and a support ring **122**. The expand ring **121** holds the wafer rings WR. The support ring **122** horizontally positions a dicing tape DT that is held by the wafer rings WR. The pushup unit **13** is disposed inside the support ring **122**.

[0032] A wafer W is adhered (attached) to the dicing tape DT, and the wafer W is divided into a plurality of dies D. The dicing tape DT is transparent to visible light. A film-like adhesive material DF called a die attach film (DAF) is attached between the wafer W and the dicing tape DT. The

adhesive material DF hardens when heated.

[0033] The wafer holding table **12** is moved in the X1-X2 and Y1-Y2 directions by a drive unit (not depicted) in order to move the dies D, which are to be picked up, to the position of the pushup unit **13**. Further, the drive unit (not depicted) causes the wafer holding table **12** to rotate the wafer rings WR within the XY plane. The pushup unit **13** is moved in the vertical direction by the drive unit (not depicted). The pushup unit **13** peels off the dies D from the dicing tape DT. The wafer holding table **12** and the pushup unit **13** form a pickup device. The pickup device may include the pickup unit **20**.

[0034] When the dies D are to be pushed up, the wafer holding table **12** lowers the expand ring **121** that holds the wafer rings WR. In this instance, the support ring **122** does not lower. Therefore, the dicing tape DT held by the wafer rings WR is stretched to increase the spacing between the dies D. This prevents interference and contact between the dies D, and thus makes it easier to separate and push up the individual dies D. The expand ring **121** and the support ring **122** are collectively referred to as an expander. The pushup unit **13** pushes up the dies D from below, thereby promoting the peeling of the dies D and improving the performance of a collet to pick up the dies D.

[0035] The wafer recognition camera **14** recognizes a pickup position of the dies D, which are to be picked up from the wafer W, and inspects the surface of the dies D.

[0036] The pickup unit **20** has a pickup head **21** and a Y drive unit **23**. The pickup head **21** is provided with a collet **22** that adsorbs and holds the peeled dies D at its tip. The pickup head **21** picks up the dies D from the wafer supply unit **10**, and places the dies D on an intermediate stage **31**. The Y drive unit **23** moves the pickup head **21** in the Y1-Y2 direction. The pickup unit **20** has various drive unit (not depicted) for raising, lowering, or rotating the pickup head **21**, or moving the pickup head **21** in the X direction.

[0037] The intermediate stage unit **30** has a stage recognition camera **34** and the intermediate stage **31** on which the dies D are placed. The stage recognition camera **34** is used for recognizing the dies D on the intermediate stage **31**. The intermediate stage **31** has a suction hole for adsorbing the dies D placed thereon. The placed dies D are temporarily held on the intermediate stage **31**. The intermediate stage **31** acts as a placement stage on which the dies D are placed, and also acts as a pickup stage from which the dies D are picked up.

[0038] The bonding unit **40** has a bond head **41**, a Y drive unit **43**, a substrate recognition camera **44**, and a bond stage **46**. The bond head **41** is provided with a collet **42** that adsorbs and holds the dies D at its tip. The Y drive unit **43** moves the bond head **41** in the Y1-Y2 direction. The substrate recognition camera **44** captures an image of a position recognition mark (not depicted) on a substrate S to recognize a bond position. Here, a plurality of product areas (hereinafter referred to as a package areas P), which will eventually become one package, are formed on the substrate S. The position recognition mark is provided for each of the package areas P. When the dies D are to be placed on the substrate S, the bond stage **46** is raised to support the substrate S from below. The bond stage **46** has a suction port (not depicted) for vacuum adsorption of the substrate S, and is capable of fastening the substrate S. The bond stage **46** has a heating portion (not depicted) for heating the substrate S. The bonding unit **40** has various driving unit (not depicted) for raising, lowering, or rotating the bond head **41** or moving the bond head **41** in the X direction.

[0039] With the above-described configuration, the bond head **41** corrects the pickup position and posture in accordance with image data obtained by the stage recognition camera **34**, and picks up the dies D from the intermediate stage **31**. Then, the bond head **41** performs bonding on the package areas P of the substrate S in accordance with the image data obtained by the substrate recognition camera **44**, or performs bonding in stack form on the dies already bonded on the package areas P of the substrate S.

[0040] The carrier unit **50** has carrying claws **51** and a carrying lane **52**. The carrying claws **51** grip and carry the substrate S. The substrate S moves in the carrying lane **52**. The substrate S moves in the X direction when a nut (not depicted) of the carrying claws **51** provided on the carrying lane **52**

is driven by a ball screw (not depicted) provided along the carrying lane **52**. With the above-described configuration, the substrate **S** moves from the substrate supply unit **60** along the carrying lane **52** to the bond position, and after bonding, moves to the substrate unloading unit **70** and delivers the substrate **S** to the substrate unloading unit **70**.

[0041] The substrate supply unit **60** removes the substrate **S**, which has been stored in a carrying jig and loaded, from the carrying jig, and supplies the substrate **S** to the carrier unit **50**. The substrate unloading unit **70** receives the substrate **S**, which has been carried by the carrier unit **50**, and stores the substrate **S** in the carrying jig.

[0042] The control unit **80** will now be described with reference to FIG. **4**. FIG. **4** is a block diagram illustrating a schematic configuration of a control system of the die bonder depicted in FIG. **1**.

[0043] The control system **8** includes, for example, the control unit (controller) **80**, a drive unit **86**, a signal unit **87**, and an optical system **88**. The control unit **80** broadly includes a control processing device **81**, a memory device **82**, an input/output device **83**, a bus line **84**, and a power supply unit **85**. The control processing device **81** is mainly formed by a CPU (Central Processing Unit). The memory device **82** includes a main memory device **82a** and an auxiliary memory device **82b**. The main memory device **82a** is formed by a RAM (Random Access Memory) that stores, for example, processing programs. The auxiliary memory device **82b** is formed by an HDD (Hard Disk Drive), an SSD (Solid State Drive), or other memory device that stores control data and image data necessary for control.

[0044] The input/output device **83** includes a monitor **83a**, a touch panel **83b**, a mouse **83c**, and an image pickup device **83d**. The monitor **83a** displays information indicating, for example, the status of a device. The touch panel **83b** inputs operator instructions. The mouse **83c** operates the monitor **83a**. The image pickup device **83d** acquires image data from the optical system **88**. The input/output device **83** further includes a motor control device **83e** and an I/O signal control device **83f**. The motor control device **83e** controls, for example, the XY table (not depicted) of the wafer supply unit **10**, the ZY drive shaft of a bond head table, and the drive unit of the pushup unit **13**. The I/O signal control device **83f** acquires or controls signals from the signal unit **87** that includes, for example, various sensors and switches and variable resistors for controlling the brightness of lighting devices. The optical system **88** includes the wafer recognition camera **14**, the stage recognition camera **34**, and the substrate recognition camera **44**. The control processing device **81** acquires necessary data through the bus line **84**, performs calculations, controls, for example, the pickup head **21**, and sends information, for example, to the monitor **83a**.

[0045] A part of a method for manufacturing semiconductor device (method for manufacturing semiconductor device) by using the die bonder **1** will now be described with reference to FIG. **5**. FIG. **5** is a flowchart illustrating a method for manufacturing the semiconductor device by using the die bonder depicted in FIG. **1**. Operations of individual sections of the die bonder **1**, which are described below, are controlled by the control unit **80**.

Wafer Loading Process (Step S1)

[0046] The wafer rings **WR** are supplied to the wafer cassette of the wafer cassette lifter **11**. The supplied wafer rings **WR** are then forwarded to the wafer holding table **12**.

Substrate Loading Process (Step S2)

[0047] The carrying jig storing the substrate **S** is supplied to the substrate supply unit **60**. In the substrate supply unit **60**, the substrate **S** is removed from the carrying jig, and fastened to the carrying claws **51**.

Pick-Up Process (Step S3)

[0048] After completion of step **S1**, the wafer holding table **12** is moved so that a desired die **D** can be picked up from the dicing tape **DT**. The wafer recognition camera **14** captures an image of the die **D**, and then the die **D** is subjected to positioning and surface inspection based on image data acquired by image capturing. Image processing is performed on the image data in order to calculate

the amount of deviation (in the X, Y, and θ directions) of the die D on the wafer holding table **12** from a die position reference point of the die bonder and perform positioning. It should be noted that a predetermined position of the wafer holding table **12** is stored in advance as an initial apparatus setting indicating the die position reference point. The surface inspection of the die D is conducted by performing image processing on the image data.

[0049] The positioned die D is peeled off from the dicing tape DT by the pushup unit **13** and the pickup head **21**. The die D peeled off from the dicing tape DT is adsorbed and held by the collet **22** disposed on the pickup head **21**, and carried to and placed on the intermediate stage **31**.

[0050] The stage recognition camera **34** captures an image of the die D on the intermediate stage **31**, and then the die D is subjected to positioning and surface inspection based on image data acquired by image capturing. Image processing is performed on the image data in order to calculate the amount of deviation (in the X, Y, and θ directions) of the die D on the intermediate stage **31** from a die position reference point of the die bonder and perform positioning. It should be noted that a predetermined position of the intermediate stage **31** is stored in advance as an initial apparatus setting indicating the die position reference point. The surface inspection of the die D is conducted by performing image processing on the image data.

[0051] The pickup head **21**, which has carried the die D to the intermediate stage, is returned to the wafer supply unit **10**. The above-described procedure is followed to peel off the next die D from the dicing tape DT. Subsequently, the dies D are peeled off one by one from the dicing tape DT by following the procedure similar to the above.

Bonding Process (Step S4)

[0052] The substrate S is carried to the bond stage **46** by the carrier unit **50**. An image of the substrate S placed on the bond stage **46** is captured by the substrate recognition camera **44**, and then the substrate S is subjected to positioning and surface inspection based on image data acquired by image capturing. An image processing is performed on the image data in order to calculate the amount of deviation (in the X, Y, and θ directions) of the substrate S from a substrate position reference point of the die bonder **1**. It should be noted that a predetermined position of the bonding unit **40** is stored in advance as an initial apparatus setting indicating the substrate position reference point. The surface inspection of the substrate S is conducted by performing image processing on the image data.

[0053] An adsorption position of the bond head **41** is corrected based on the amount of deviation of the die D on the intermediate stage **31**, which has been calculated in step S3, and then the die D is adsorbed by the collet **42**. The bond head **41**, which has adsorbed the die D from the intermediate stage **31**, bonds the die D to a predetermined location on the substrate S supported by the bond stage **46**. In this instance, the predetermined location of the substrate S is a package area P of the substrate S, or an area where an element is already placed and another element is to be bonded in addition to that, or a bonding area for an element to be bonded in stack form. An image of the die D bonded to the substrate S is captured by the substrate recognition camera **44**, and then inspection is performed, for example, to determine whether the die D has been bonded at a desired position based on the image data acquired by image capturing.

[0054] The bond head **41**, which has bonded the die D to the substrate S, is returned to the intermediate stage **31**. The above-described procedure is followed to pick up the next die D from the intermediate stage **31** and bond the next die D to the substrate S. The above is repeated to bond the dies D to all the package areas P of the substrate S.

Substrate Unloading Process (Step S5)

[0055] The substrate S, to which the dies D are bonded, is carried to the substrate unloading unit **70**. In the substrate unloading unit **70**, the substrate S is removed from the carrying claws **51** and stored in the carrying jig. The carrying jig storing the substrate S is unloaded from the die bonder **1**.

[0056] As described above, the dies D are mounted on the substrate S and unloaded from the die bonder **1**. Subsequently, for example, the carrying jig storing the substrate S, on which the dies D

are mounted, is carried to a wire bonding process, and then electrodes of the dies D are electrically connected to electrodes of the substrate S, for example, via Au wires. The substrate S is then carried to a molding process, where the die D and the Au wires are sealed with a mold resin (not depicted) to complete a semiconductor package.

[0057] The pushup unit **13** will now be outlined with reference to FIGS. **6** and **7**. FIG. **6** is a front view of a pushup unit in the embodiment. FIG. **7** is a schematic diagram illustrating the pushup unit depicted in FIG. **6**.

[0058] As depicted in FIG. **6**, the pushup unit **13** includes a head section **100**, a mechanism section **200**, and a drive section **300**. As depicted in FIG. **7**, the head section **100** includes a pushup block section BLK that has a plurality of blocks B1 to B8. The mechanism section **200** drives the blocks B1 to B8 in the vertical direction. The mechanism section **200** includes an upper plate section **210** and a lower block section **220**. The upper plate section **210** acts as a first mechanism section or a second mechanism section. The lower block section **220** acts as the first mechanism section or the second mechanism section. The drive section **300** generates a vertical motion, and transmits the vertical motion to the lower block section **220**. The lower block section **220** transmits the vertical motion, which is generated by the drive section **300**, to the upper plate section **210**. The upper plate section **210** transmits a vertical motion generated by the lower block section **220** to the head section **100**.

[0059] The mechanism section **200** may be formed by the upper plate section **210** and the drive section **300**. In this case, the drive section **300** generates a vertical motion, and transmits the vertical motion to the upper plate section **210**. The upper plate section **210** transmits a vertical motion generated by the drive section **300** to the head section **100**. The mechanism section **200** may be formed by the lower block section **220** and the drive section **300**. In such an instance, the drive section **300** generates a vertical motion, and transmits the vertical motion to the lower block section **220**. The lower block section **220** transmits a vertical motion generated by the drive section **300** to the head section **100**.

[0060] As depicted in FIG. **7**, the blocks B1 to B8 can be independently moved up and down by the mechanism section **200** and by drive shafts ND1 to ND8 of the drive section **300**. Each of the drive shafts ND1 to ND8 is formed by a later-described motor M (see FIG. **16**) and plunger mechanism in order to vertically move the blocks B1 to B8. The plunger mechanism includes the upper plate section **210**, the lower block section **220**, and the drive section **300** excluding the motor M.

[0061] The head section **100** will be described with reference to FIGS. **6** and **8**. FIG. **8** is a top view of a head section depicted in FIG. **6**.

[0062] As depicted in FIG. **8**, the head section **100** includes a dome **110** and the pushup block section BLK, which is disposed in the dome **110**. The dome **110** includes a cylindrical member **111**, a disk-shaped member **112**, and another cylindrical member **113**. The cylindrical members **111**, **113** are disposed on the upper end of the cylindrical member **111**. An opening is provided in the center of the disk-shaped member **112** so that the pushup block section BLK is able to move up and down through this opening. The periphery of the disk-shaped member **112** is provided with a plurality of suction ports **112a** and a plurality of grooves **112b**. The plurality of grooves **112b** connect the plurality of suction ports **112a**. The inside of the suction ports **112a** is depressurized by a suction mechanism (not depicted) when the pushup unit **13** is to be raised to bring its upper surface into contact with the back surface of the dicing tape DT. In this instance, the back surface of the dicing tape DT is sucked downward and brought into close contact with the upper surface of the dome **110** (disk-shaped member **112**). As depicted in FIG. **6**, an annular member **114** is disposed on the outside of the cylindrical member **113**. The dome **110** is fastened to the upper plate section **210** by the annular member **114**.

[0063] The blocks B1 to B8 of the pushup block section BLK push the dicing tape DT upward. The seven outer blocks B1 to B7 have a hollow columnar shape (e.g., a square cylindrical shape), and are provided with openings that penetrate in the Z1-22 direction and have the same shape as the

outer shape of an internally positioned block. The innermost block **B8** has a solid columnar shape (e.g., a square columnar shape). The block **B2** is placed inside the block **B1**, which is the largest in size, the block **B3** is placed inside the block **B2**, and the block **B4** is placed inside the block **B3**. The block **B5** is placed inside the block **B4**, the block **B6** is placed inside the block **B5**, the block **B7** is placed inside the block **B6**, and the block **B8**, which is the smallest in size, is placed further inside.

[0064] It is preferable that the outermost block **B1**, which is the largest in size among the eight blocks **B1** to **B8**, be slightly smaller in size than the outer circumference of the die **D** to be peeled off. As a result, the corners of the upper surface of the block **B1** are positioned slightly inside the outer edge of the die **D**. Consequently, the force for peeling the die **D** and the dicing tape **DT** from each other can be concentrated at a point (the outermost periphery of the die **D**) that serves as the starting point for peeling the two apart. Here, the outer periphery of the die **D** protruding outward from the end of the outermost block **B1** is called an overhang (**OH**).

[0065] The upper plate section **210** of the mechanism section **200** will now be described with reference to FIGS. **9** to **11**. FIG. **9** is a longitudinal cross-sectional view of an upper plate section depicted in FIG. **6**. FIG. **10** is a transverse cross-sectional view of the upper plate section taken along line C-C in FIG. **9**. FIG. **11** is a transverse cross-sectional view of the upper plate section taken along line E-E in FIG. **9**.

[0066] As depicted in FIG. **9**, the upper plate section **210** includes an upper rod **UR**, an upper plate **UP**, and a dome. The dome covers the upper rod **UR** and the upper plate **UP**. The dome is formed by a cylindrical member **211** and a disk-shaped member **212**. The disk-shaped member **212** is disposed on the upper end of the cylindrical member **211**. An annular member **213** is disposed on the outside of the cylindrical member **211**. The cylindrical member **211** is fastened to the lower block section **220** by the annular member **213**. An opening **212a** is provided in the center of the disk-shaped member **212** so that the upper rod **UR** is able to move up and down through this opening **212a**. The upper rod **UR** includes rods **UR1** to **UR7**. The upper plate **UP** includes plates **UP1** to **UP7**. Each of the rods **UR1** to **UR7** is referred to as a first rod, a second rod or a third rod.

[0067] Each of the rods **UR1** to **UR7** is disposed on the upper surface side of the corresponding one of the plates **UP1** to **UP7**, and extended in the **Z** direction. The rods **UR1** to **UR7** are fastened at one end to the plates **UP1** to **UP7** and at the other end to the blocks **B1** to **B7**. It is preferable that the outermost block **B1** use the top plate **UP1**. As a result, the outer block **B1**, which has a large outer shape and needs to be rigid, can be pushed up by the rod **UR1**, which is the shortest, highly rigid pin and used as a plunger.

[0068] As depicted in FIG. **10**, four pieces of the rod **UR1** are provided, and three pieces each of the rods **UR2** to **UR7** are provided. This allows the block **B1** to be supported at four points, and each of the blocks **B2** to **B7** to be supported at three points, thereby providing increased rigidity between the block section **BLK** and the upper plate **UP** and providing increased parallelism during ascent and descent. Three or more of each of the rods **UR1** to **UR7** may be provided. It is preferable that a plurality of rods forming the rod **UR1** be arranged concentrically at equal intervals. It is preferable that a plurality of rods forming each of the rods **UR2** to **UR7** be also arranged concentrically at equal intervals. This increases the parallelism of the blocks **B1** to **B7** during ascent and descent. The rods **UR1** to **UR7** transmit the vertical motions of the plates **UP1** to **UP7** to the blocks **B1** to **B7**.

[0069] Additionally, the mechanism section **200** has a rod **LR8** that passes through the upper plate section **210**. The rod **LR8** is fastened at one end to a block **LB8** of a later-described lower block **LB** and at the other end to the block **B8** of the block section **BLK**. Only one piece of the rod **LR8** is provided. It is preferable that the rod **LR8** be larger in diameter than the rods **UR1** to **UR7**.

[0070] The plates **UP1** to **UP7** are arranged at predetermined intervals in the vertical direction (**Z** direction). The plates **UP1** to **UP7** are disk-shaped members. Further, the plates **UP1** to **UP7** have

operational clearance from the inner wall of the dome (cylindrical member **211**). The above-described configuration makes it possible to dispose many rods and maintain their parallel and vertical orientation during operation. The number of pieces of the upper plate UP is one less than the number of blocks in the block section BLK. Alternatively, however, they may be the same (eight). However, when the innermost block **B8** is pushed up by the rod **LR8** on an as-is basis (without using a plate), the mechanism can be simplified because the parallelism of the innermost block **B8** can be easily maintained due to its small outer shape.

[0071] The plates **UP1** to **UP6** have through-holes PTH through which the rods **UR2** to **UR7** pass. The rods **UR2** to **UR7** are to be connected to the plates **UP2** to **UP7**, which are positioned below the plates **UP2** to **UP7**, respectively. Further, the plates **UP2** to **UP7** have through-holes PTH through which rods **LR1** to **LR6** pass. The rods **LR1** to **LR6** are to be connected to later-described blocks **LB1** to **LB7**, which are used for connecting, respectively, to the upper plates **UP1** to **UP6** positioned above. Then, the plates **UP1** to **UP7** have through-holes PTH through which the rod **LR8** passes. It is preferable that the through-holes PTH and the rods **UR2** to **UR7**, **LR8** have the same shape when viewed from above. Clearance is provided to allow driving. It should be noted that the through-holes PTH and the rods **UR2** to **UR7**, **LR8** do not necessarily have the same shape. This allows the dome to accommodate a link mechanism that transmits many driving forces from the lower block section **220** to the pushup block section BLK.

[0072] The lower block section **220** of the mechanism section **200** will be described with reference to FIG. **6**, **12** or **13**. FIG. **12** is a longitudinal cross-sectional view of a lower block section depicted in FIG. **6**. FIG. **13** is a transverse cross-sectional view of the lower block section taken along line F-F in FIG. **12**.

[0073] As depicted in FIG. **12**, the lower block section **220** includes a lower rod LR, a lower block LB, and a housing that covers the lower rod LR and the lower block LB. The housing is formed by members **221** that are provided on a back section and both side sections. The lower rod LR includes the rods **LR1** to **LR8**. The lower block LB includes the block **LB1** to **LB8**. The each of the rods **LR1** to **LR8** is referred to as a first rod, a second rod or a third rod.

[0074] Each of the rods **LR1** to **LR8** is disposed on the upper surface side of the corresponding one of the blocks **LB1** to **LB8**, and extended in the Z direction. The rods **LR1** to **LR8** are fastened at one end to the blocks **LB1** to **LB8** and at the other end to the plates **UP1** to **UP7**. The other end of the rod **LR8** is fixed to the block **B8**.

[0075] As depicted in FIG. **13**, three pieces each of the rods **LR1** to **LR7** are provided. This allows each of the blocks **B1** to **B7** to be supported at three points, thereby providing increased rigidity between the upper plate UP and the lower block LB and providing increased parallelism during ascent and descent. Four or more of each of the rods **LR1** to **LR7** may be provided. It is preferable that a plurality of rods forming the rod **LR1** be arranged concentrically at equal intervals. It is preferable that a plurality of rods forming each of the rods **LR2** to **LR7** be arranged concentrically at equal intervals. Further, it is preferable that a plurality of rods forming the rod **LR1** be arranged at equal angular intervals. It is preferable that a plurality of rods forming each of the rods **LR2** to **LR7** be arranged at equal angular intervals. This increases the parallelism of the blocks **B1** to **B7** during ascent and descent. The rods **LR1** to **LR7** transmit the vertical motions of the blocks **LB1** to **LB7** to the plates **UP1** to **UP7**. The rod **LR8** transmits the vertical motions of the block **LB8** to **UP7** to the block **B8**.

[0076] The blocks **LB1** to **LB8** are arranged in the vertical direction. The blocks **LB1** to **LB8** are circular, rectangular, or polygonal members. The blocks **LB1** to **LB8** are formed to be thicker than the plates **UP1** to **UP7**. The blocks **LB1** to **LB7** have through-holes BTH through which the rods **LR2** to **LR8** pass. The rods **LR2** to **LR8** are to be connected to the blocks **LB2** to **LB8**, which are positioned below the blocks **LB1** to **LB7**, respectively. It is preferable that the through-holes BTH and the rods **LR2** to **LR8** have the same shape when viewed from above. Clearance is provided to allow driving on an independent basis. It should be noted that the through-holes PTH and the rods

UR2 to UR7, LR8 do not necessarily have the same shape.

[0077] The drive section **300** will be described with reference to FIGS. **14** to **17**. FIG. **14** is a top view illustrating an example of layout of the drive section depicted in FIG. **6**. FIG. **15** is a side view illustrating an example layout of the drive section depicted in FIG. **6**. FIG. **16** is a diagram illustrating an example of the configuration of the drive section depicted in FIG. **15**. FIG. **17** is a diagram illustrating another example of the configuration of the drive section depicted in FIG. **15**.

[0078] The drive section **300** includes drive sections DU1 to DU8. The drive sections DU1 to DU8 are arranged around the lower block section **220** and fastened to the members **221**. The drive sections DU1, DU7, DU8 are disposed on the X2 side of the lower block section **220**. The drive sections DU2, DU5, DU6 are disposed on the X1 side of the lower block section **220**. The drive sections DU3 and DU4 are disposed on the Y1 side of the lower block section **220**. The drive sections DU1 and DU2 are disposed on the upper side of the lower block section **220**. The drive sections DU5 to DU8 are disposed on the lower side of the lower block section **220**.

[0079] The drive sections DU1 to DU8 respectively drive the blocks LB1 to LB8 in the vertical direction. Each of the drive sections DU1 to DU8 includes a motor M and a drive output section PM. The drive output section PM converts the rotation of the motor M into a vertical motion.

[0080] The drive output section PM includes a ball screw BS. The ball screw BS is formed by a screw shaft SS, which is connected to the motor M, and a nut section NU, which is threadably engaged with the screw shaft SS. The nut section NU is screwed or otherwise fastened to the lower block LB. The motor M is disposed so as to extend its output shaft in the vertical direction. That is to say, the screw shaft SS of the ball screw BS is disposed so as to extend in the vertical direction.

[0081] The motor M may be disposed with its output shaft (motor shaft) facing upward as depicted in FIG. **16**, or may be disposed with its output shaft facing downward as depicted in FIG. **17**. That is to say, the drive output section PM may be disposed above the motor M, or may be disposed below the motor M. The motors M of the drive sections DU1, DU2, DU5 to DU8 are disposed as depicted in FIG. **16**. The motors M of the drive sections DU3, DU4 are disposed as depicted in FIG. **17**. This allows the motor M to be compactly accommodated, and provides flexibility in arrangement.

[0082] Further, the lower block section **220** receives the thrust of the drive output section PM by using the side surface or upper or lower surface of the lower block LB, and transmits motive power to the lower rod LR. This provides improved flexibility in arrangement of the drive output section PM and of a motive power source (motors M) in a case where many motive power sources (motors M) are provided.

[0083] The present embodiment provides at least one of the following advantageous effects (a) to (g). [0084] (a) It is possible to increase the number of blocks in the pushup block section BLK.

[0085] (b) Since the peel area per block (block width) can be reduced due to advantageous effect (a), it is possible to reduce the stress on the dies during die peeling. [0086] (c) Even in a case where the dies are increased in size, the overhang can be reduced due to advantageous effect (a). [0087]

(d) Since the outer periphery of a die can be peeled off at a low pushup height to position an outermost peripheral pushup block lower than a total pushup height due to advantageous effect (c), it is possible to reduce the deformation of surrounding dies. This shortens the distance between the end face of the outermost peripheral pushup block and the end face of a pickup die. As a result, the surrounding dies are less likely to deform. [0088] (e) Product defects can be reduced due to advantageous effect (d). [0089] (f) Thin dies can be processed in a stable manner due to advantageous effect (d). [0090] (g) Since the upper plate section **210** includes a disk-shaped member and a rod, it is possible to suppress an increase in a dome diameter even if the number of blocks in the pushup block section BLK is increased. This makes it possible to maintain compatibility with the pushup unit having a smaller number of blocks. This will be explained below.

[0091] As described above, the wafer holding table **12** is moved in the X and Y directions by the

XY table so that the die D to be picked up is moved to the position of the pushup unit **13**. However, in a case where the die D to be picked up is in the vicinity of the wafer rings WR, the pushup unit **13** is positioned near the members of the wafer holding table **12**, such as the support ring **122** and the XY table. If the dome **110** of the pushup unit **13** and the dome of the upper plate section **210** are increased in size, these members of the wafer holding table **12** become an obstacle. As a result, the die D to be picked up is unable to move to the position of the pushup unit **13**. That is to say, the area on the wafer that permits die pickup (pickup area) is reduced. Consequently, in a case where the configuration of the wafer holding table **12** remains unchanged, the pushup unit **13** cannot be made larger in size than a predetermined value. It should be noted that, if the number of blocks is large in a case where a second unit **13b** in Japanese Unexamined Patent Application Publication No. 2017-224640 includes a plurality of circular tubular blocks, the diameter of the second unit **13b** increases due to machining accuracy.

[0092] The present disclosure made by the present inventors has been specifically described above based on an embodiment. However, it is obvious that the present disclosure is not limited to the foregoing embodiment and can be variously modified.

[0093] For example, the foregoing embodiment has been described with reference to an example in which the pushup block section BLK includes eight blocks B1 to B8. However, the number of blocks included in the pushup block section BLK may be larger or smaller than eight. The number of blocks in the pushup block section BLK should be two or more.

[0094] In a case where the number of blocks in the pushup block section BLK is not eight, the configurations of the head section **100** and mechanism section **200** are to be changed. The numbers of upper plates, lower blocks, and motors are to be changed in accordance with the number of blocks. For example, in a case where the pushup block section BLK includes two blocks, one upper plate, two lower blocks, and two motors are to be provided.

[0095] In a case where the number of blocks in the pushup block section BLK is smaller than eight, only the configuration of the head section **100** is to be changed, and the configuration of the mechanism section **200** need not be changed. In this case, there is an upper rod that is not connected to the block section BLK; however, the motor for moving such the upper rod is to remain inoperative.

[0096] The foregoing embodiment has been described with reference to an example in which clearance is provided between the through-hole BTH in the lower block LB and the lower rod LR so that they can be driven independently. However, for example, a bearing may alternatively be provided to permit independent driving. When the above alternative is adopted, the plunger (rod) for transmitting motive power between the plates (surfaces) is kept perpendicular and can be operated with high precision.

[0097] The foregoing embodiment has been described with reference to an example in which the upper plate UP is disk-shaped. However, the upper plate UP may alternatively be shaped like a polygon so that the inner surface of the dome comes into contact with a plurality of (three or more) points or faces. Further, bearings or pulleys may be provided on the sides of the polygon. Adopting the above alternative makes it possible to maintain the parallelism of a plurality of upper plates UP.

[0098] The foregoing embodiment has been described with reference to an example in which the die attach film is used. However, an alternative is to provide a preform unit for applying an adhesive to the substrate without using the die attach film.

[0099] The die bonder described in conjunction with the foregoing embodiment is configured such that the pickup head picks up the dies from the wafer supply unit, places the dies on the intermediate stage, and allows a bonding head to bond the dies placed on the intermediate stage to the substrate. However, the present disclosure is not limited to the above-mentioned configuration. The present disclosure can also be applied to a die bonding device that picks up dies from a die supply unit.

[0100] For example, the present disclosure is also applicable to a die bonder that has neither an

intermediate stage nor a pickup head and allows the bonding head to bond the dies in the wafer supply unit to the substrate.

[0101] Further, the present disclosure is also applicable to a flip chip bonder that does not have an intermediate stage, picks up a die from a wafer supply unit, rotates a die pickup head upward to deliver the die to a bonding head, and allows the bonding head to bond the dies to a substrate.

[0102] In the foregoing embodiment, a die bonder has been described as an example. However, the present disclosure is also applicable to a semiconductor manufacturing apparatus that places picked-up dies on a tray.

Claims

1. A semiconductor manufacturing apparatus comprising: a pushup unit that includes a head section having a plurality of blocks, a mechanism section for moving each of the plurality of blocks up and down on an independent basis, and a drive section that includes a motor and a drive output section for converting the motive power of the motor to a vertical motion; and a head that has a collet adsorbing a die, wherein the mechanism section includes a first mechanism section having a first member that has an upper surface, a lower surface opposite the upper surface, and a through-hole penetrating between the upper surface and the lower surface, a first rod that is connected to the upper surface of the first member to convey the vertical motion of the first member to the blocks, and a second rod that is extended from below through the through-hole to convey the vertical motion to the blocks.
2. The semiconductor manufacturing apparatus according to claim 1, wherein the mechanism section further includes a second mechanism section having a third rod that is connected to the lower surface of the first member, and a second member that has an upper surface, a lower surface opposite the upper surface, and a through-hole penetrating between the upper surface and the lower surface, and is configured to connect the third rod to the upper surface.
3. The semiconductor manufacturing apparatus according to claim 1, wherein the mechanism section further includes a second mechanism section having a second member that has an upper surface, a lower surface opposite the upper surface, and a through-hole penetrating between the upper surface and the lower surface, and is configured to connect the first rod to the lower surface, and a third rod that is connected to the upper surface of the second member.
4. The semiconductor manufacturing apparatus according to claim 2, wherein a plurality of pieces of the first member of the first mechanism section are arranged in the vertical direction; wherein a plurality of pieces of the second member of the second mechanism section are arranged in the vertical direction; wherein the drive section has a plurality of pieces of the motor and a plurality of pieces of the drive output section; and wherein the number of the first members of the first mechanism section is equal to or smaller than the number of the second members of the second mechanism section.
5. The semiconductor manufacturing apparatus according to claim 2, wherein a plurality of pieces of the first member of the first mechanism section are arranged in the vertical direction; wherein a plurality of pieces of the second member of the second mechanism section are arranged in the vertical direction; wherein the drive section has a plurality of pieces of the motor and a plurality of pieces of the drive output section; wherein the number of the first members of the first mechanism section is equal to or smaller than the number of the blocks; wherein the number of the second members of the second mechanism section is equal to the number of the drive output sections; and wherein the number of the drive output sections is equal to the number of the motors.
6. The semiconductor manufacturing apparatus according to claim 4, wherein at least three pieces of the first rods are disposed on the upper surface of each of the plurality of the first members of the first mechanism section; wherein, one piece of the second rod is disposed on the upper surface of one piece of the plurality of second members of the second mechanism section; and wherein at

least three pieces of the third rod are disposed on the upper surface of each of the plurality of the other second members of the second mechanism section.

7. The semiconductor manufacturing apparatus according to claim 5, wherein at least three pieces of the first rods are disposed on the upper surface of each of the plurality of the first members of the first mechanism section; wherein, one piece of the second rod is disposed on the upper surface of one piece of the plurality of second members of the second mechanism section; and wherein at least three pieces of the third rod are disposed on the upper surface of each of the plurality of the other second members of the second mechanism section.

8. The semiconductor manufacturing apparatus according to claim 6, wherein the first rods are arranged concentrically at equal intervals; and wherein the third rods are arranged concentrically at equal intervals.

9. The semiconductor manufacturing apparatus according to claim 4, wherein the first mechanism section is accommodated in a cylindrical housing; and wherein the first members of the first mechanism section are circular or polygonal in shape when viewed from above, and clearance is provided to allow driving between the first members and the inside diameter of the housing.

10. The semiconductor manufacturing apparatus according to claim 5, wherein the first mechanism section is accommodated in a cylindrical housing; and wherein the first members of the first mechanism section are circular or polygonal in shape when viewed from above, and clearance is provided to allow driving between the first members and the inside diameter of the housing.

11. The semiconductor manufacturing apparatus according to claim 2, wherein the second members of the second mechanism section are circular, rectangular, or polygonal in shape when viewed from above.

12. The semiconductor manufacturing apparatus according to claim 4, wherein the outer ones of the plurality of the blocks are pushed up by the first rods disposed on the upper ones of the plurality of the first members of the first mechanism section.

13. The semiconductor manufacturing apparatus according to claim 5, wherein the outer ones of the plurality of the blocks are pushed up by the first rods disposed on the upper ones of the plurality of the first members of the first mechanism section.

14. The semiconductor manufacturing apparatus according to claim 4, wherein the second rod is configured to push up the innermost one of the plurality of the blocks without being connected to the plurality of the first members of the first mechanism section.

15. The semiconductor manufacturing apparatus according to claim 5, wherein the second rod is configured to push up the innermost one of the plurality of the blocks without being connected to the plurality of the first members of the first mechanism section.

16. The semiconductor manufacturing apparatus according to claim 5, wherein each of the plurality of the drive output sections is fastened to the plurality of the second members of the second mechanism section.

17. The semiconductor manufacturing apparatus according to claim 12, wherein the drive section is disposed around, above, or below the second mechanism section.

18. The semiconductor manufacturing apparatus according to claim 13, wherein the plurality of the motor shafts of the motors are oriented upward or downward.

19. A pushup unit comprising: a head section having a plurality of blocks, a mechanism section for moving each of a plurality of blocks up and down on an independent basis, a motor, and a drive section having a drive output section for converting the motive power of the motor to a vertical motion, wherein the mechanism section includes a first mechanism section having a member that has an upper surface, a lower surface opposite the upper surface, and a through-hole penetrating between the upper surface and the lower surface, a first rod that is connected to the upper surface of the member to convey the vertical motion of the member to the blocks, and a second rod that is extended from below through the through-hole to convey the vertical motion to the blocks.

20. A method for manufacturing semiconductor device, comprising the steps of: loading a wafer

ring holding a dicing tape into the semiconductor manufacturing apparatus according to claims 1;
and picking up a die attached to the dicing tape.
